SINGAPORE DIMENSIONS • GRADE 1A • CHAPTER 3

Addition as Putting Together & Part-Part-Whole

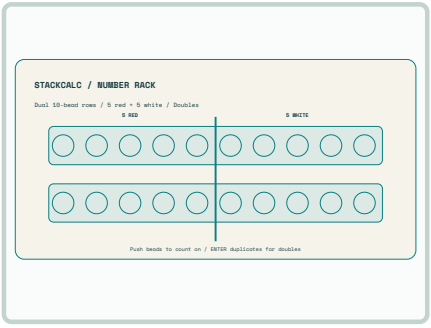
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.1.OA.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Rack

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Push two bead groups on the number rack together to form a combined quantity.

Sketch Model on Paper:
Draw a part-part-whole bar model with segments labeled.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	6	X: 6	Key 1st operand
2.	ENTER	Y: 6 | X: 6	Stack lift copies X
3.	3	Y: 6 | X: 3	Key 2nd operand
4.	+	X: 9	Drop stack & compute

EXPECTED LCD X: 9 apples

1 4 red blocks and 5 blue blocks. How many blocks in all?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 7 birds on branch, 2 more land. How many birds?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 3 toy cars in box, 5 on rug. Total cars?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 8 green pencils and 2 yellow pencils. Total pencils?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 3

Addition as Putting Together & Part-Part-Whole • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Math Domino Studio & Number Rack

CAD Blueprint: number_rack

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/number_rack_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

White & Signal Red Beads (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**4 red blocks and 5 blue blocks. How many blo** Standard: CCSS.MATH.1.OA.A.1	9	[4] [ENTER] [5] [+]
2	**7 birds on branch, 2 more land. How many bir** Standard: CCSS.MATH.1.OA.A.1	9	[7] [ENTER] [2] [+]
3	**3 toy cars in box, 5 on rug. Total cars?** Standard: CCSS.MATH.1.OA.A.1	8	[3] [ENTER] [5] [+]
4	**8 green pencils and 2 yellow pencils. Total** Standard: CCSS.MATH.1.OA.A.1	10	[8] [ENTER] [2] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Remind students that ENTER is only pressed once between the two operands.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.